Learning Objective 5.1: I can create a GNU Radio out-of-tree (OOT) module and add a custom Python block to it.

Learning Objective 5.2: I can implement a custom signal-processing block from scratch and integrate it into a GNU Radio flowgraph.

Show your work. Write a **Documentation** line at the end (**None** if you did not collaborate).

1. **The files of a module.** A cadet runs `gr_modtool newmod beacon` and then `gr_modtool add -t sync -l python tone_finder` in the new directory.

   (a) Name the file the cadet edits for each job: (i) the signal processing, (ii) the label and parameters GRC shows, and (iii) the unit test that `ctest` runs.

   (b) The cadet builds and installs the module with GNU Radio 3.10.9. `ctest` passes, but a flowgraph that uses the block stops with

   ```
   ModuleNotFoundError: No module named 'gnuradio.beacon.tone_finder'
   ```

   Explain the cause and the fix.

   (c) After a correct install, GRC's block library still has no `tone_finder`. Give two possible causes.

2. **Choosing a block type.** For each block below, choose sync, decimator, interpolator, or general, and give the ratio of outputs to inputs.

   (a) A block that converts each float sample to dB.

   (b) A block that outputs the peak magnitude of each run of 100 complex samples.

   (c) A block that holds each bit for 8 samples.

   (d) A squelch that passes only the samples whose power is above a threshold and drops the rest.

   (e) The Energy Detector could have been a general block. Explain in words what code a decimator lets the programmer leave out.

3. **Levels.** A cadet runs the Energy Detector on a keyed tone of amplitude 1 at 48 kHz, with GNU Radio's Noise Source at amplitude 0.25, and averages 64 samples.

   (a) Find the power output, in dB, with the carrier off and with it on.

   off:

   on:

   (b) Choose a threshold halfway between the two levels in dB.

   (c) How long does each output cover, and how many outputs does the block make per second?

   (d) Averaging N samples divides the spread of the noise power by $\sqrt{N}$. Find the floor's spread as a fraction of its mean, and in dB above and below it.

   (e) Suggest a hysteresis and explain the choice in words.

4. **Hysteresis by hand.** The Energy Detector, with its key starting at 0, receives these ten power values in dB:

$$-25,\ -14,\ -16,\ -14,\ -16,\ -14,\ -21,\ -16,\ -14,\ -22$$

(a) Write the ten key outputs for a threshold of -15 dB and a hysteresis of 4 dB, and count the times the key changes.

(b) Repeat for a hysteresis of 0 dB.

(c) The values -14 and -16 dB are noise on a carrier that stays on. Explain in words which key is correct, and what a Morse decoder would make of the other.

5. **A unit test.** A test feeds the Energy Detector, with 4 samples averaged, a threshold of -10 dB, and a hysteresis of 3 dB, these twelve complex samples:

$$1,\ 1,\ 1,\ 1,\ \ 0.1,\ 0.1,\ 0.1,\ 0.1,\ \ 1,\ 1,\ 0,\ 0$$

(a) How many outputs does each port make? Give the power outputs in dB.

(b) Give the three key outputs.

(c) The last output averages a run where the carrier switched off halfway through. Explain in words what that does to the timing of a key edge, and how the number of samples averaged sets the size of the effect.

Documentation: _____